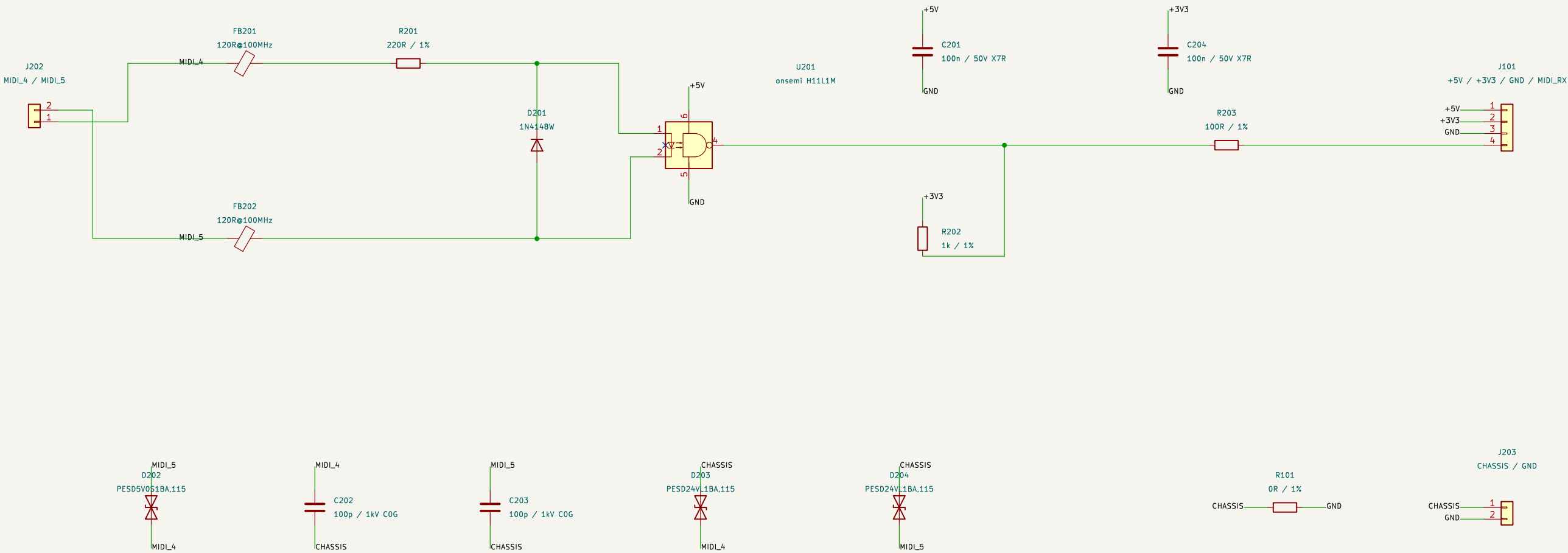


Isolated MIDI input / 3.3 V UART

REV M1 | EXTERNAL HOST + REGULATED POWER | NO OTHER MODULE REQUIRED



DIN panel socket: contact 4 -> J202.1; contact 5 -> J202.2. Pins 1/2/3 and shell unconnected.
Insulate DIN shell. J203.1 bonds the enclosure; R101 returns ESD to local GND.
Optocoupler LED current -> UART low. The output pull-up is 3.3 V, never 5 V.
Functional ground-loop isolation; transient clamps limit common-mode range. Not a safety isolator.

+3V3 +5V CHASSIS GND

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_MIDI.kicad_sch

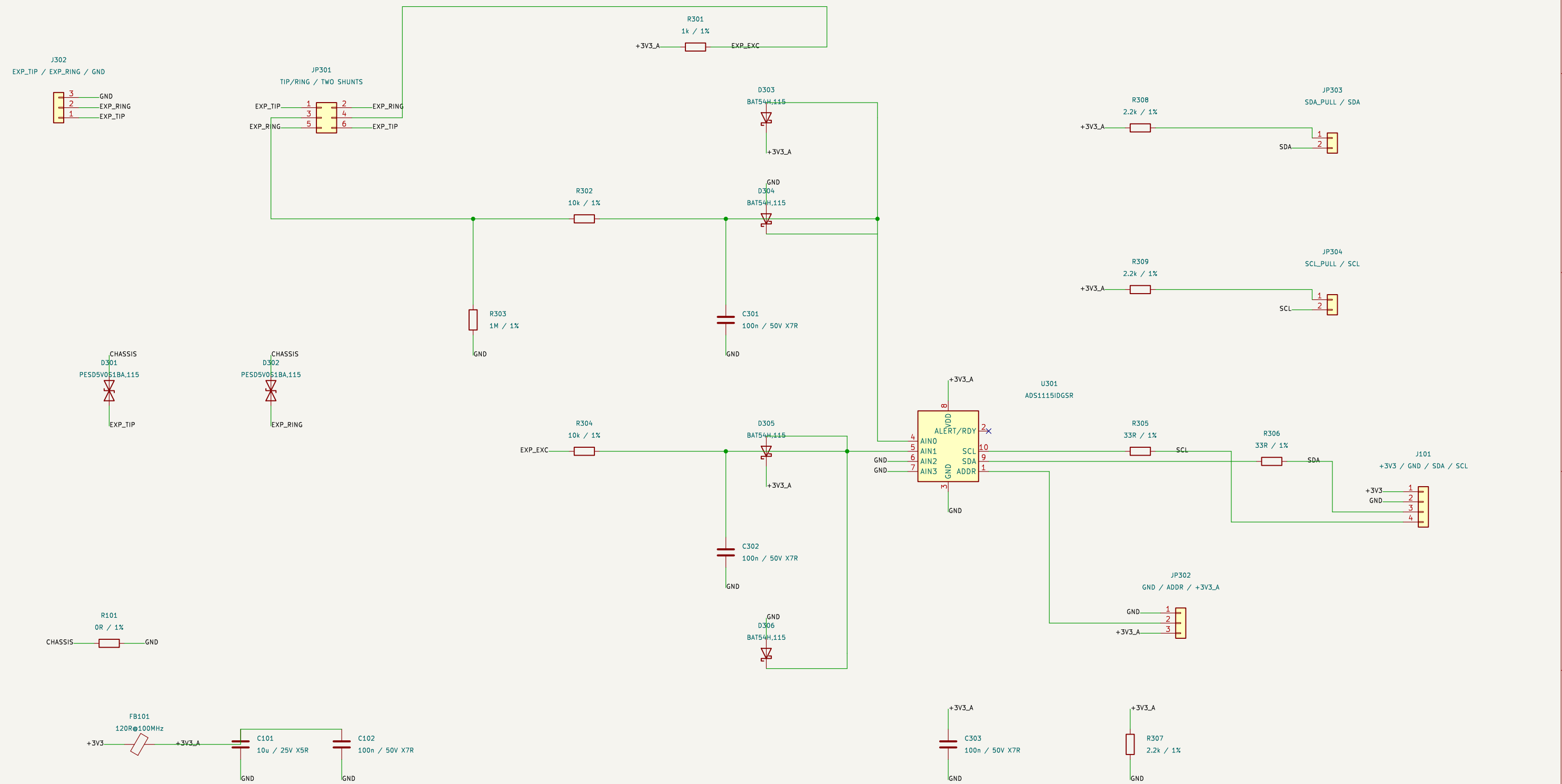
Title: Isolated MIDI input / 3.3 V UART

Size: A3 Date: 2026-10-02
KiCad E.D.A. 9.0.2

Rev: M1
Id: 1/1

Passive expression pedal / I2C ADC

REV M1 | EXTERNAL HOST + REGULATED POWER | NO OTHER MODULE REQUIRED



Passive 10k–100k pot only. Sleeve = GND. JP301: both shunts 1–3 / 2–4 for tip wiper; 3–5 / 4–6 for ring wiper.
JP302: 1–2 = 0x48; 2–3 = 0x49. Never leave ADDR floating. JP303/304 enable local 2.2k I2C pull-ups.
Remove BOTH pull-up shunts when host already has pull-ups (Raspberry Pi). Start at 100 kHz I2C.
ADS1115: +/-4.096 V, 128 SPS. Read AIN0 and AIN1; normalize AIN0/AIN1, calibrate endpoints and smooth.

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Expression.kicad_sch

Title: Passive expression pedal / I2C ADC

Size: A3 Date: 2026–10–02

Rev: M1

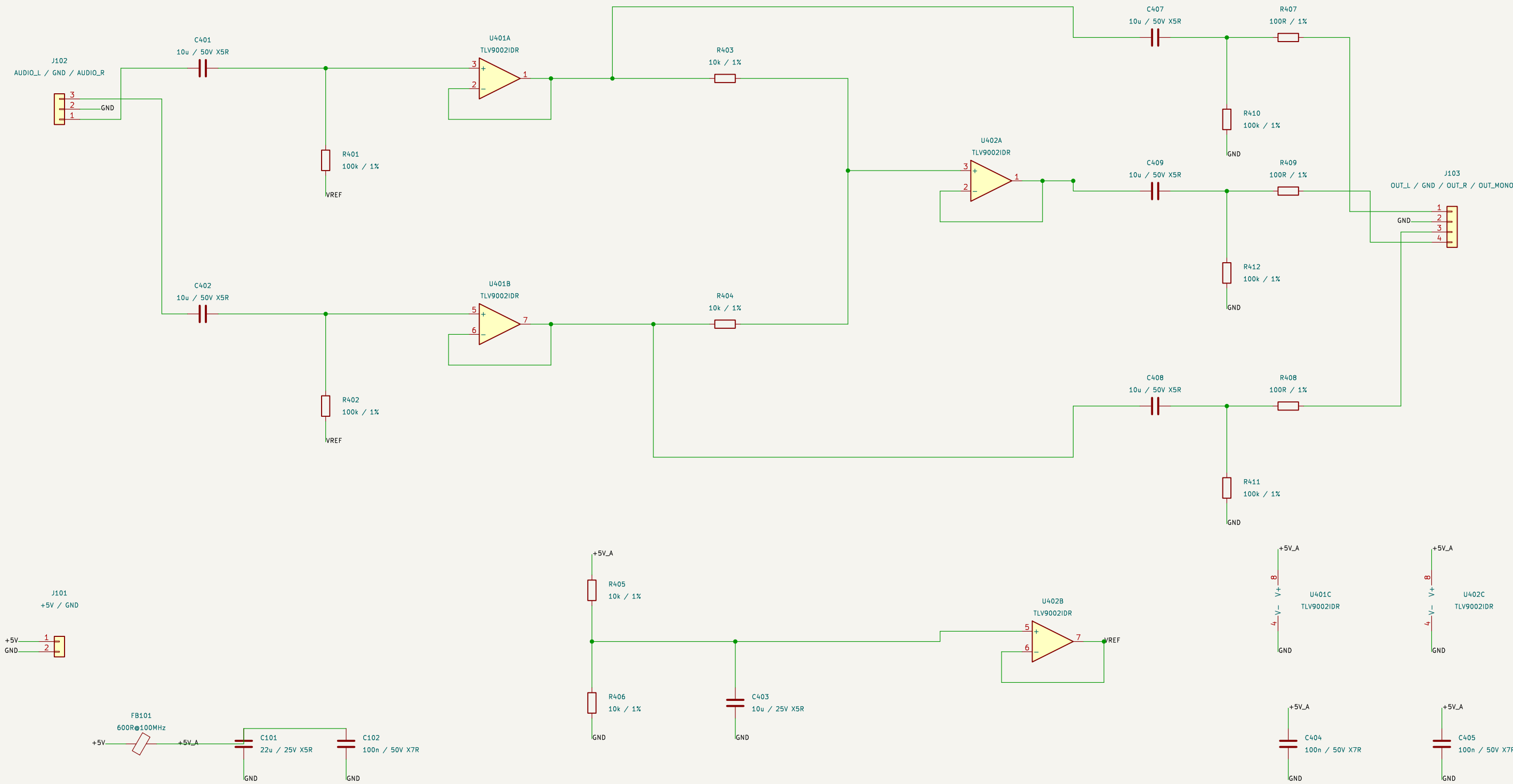
KiCad E.D.A. 9.0.2

Id: 1/1

ARDOR / INDEPENDENT FUNCTION MODULE

Stereo buffer / mono average / AC outputs

REV M1 | EXTERNAL HOST + REGULATED POWER | NO OTHER MODULE REQUIRED



Line-level input ≤ 1 Vrms/channel. Output loads $\geq 10k$. Mono = (L + R)/2, unity buffers.
All three outputs AC coupled; no 2.5 V bias leaves this board. For mono input, feed both channels
or accept -6 dB when using one channel. Harness connectors are internal; add jack-entry ESD protection
for exposed ports. This board is neither an instrument preamp nor an ADC, headphone or power amplifier.

Engineering prototype – bench validation required

Ardor

Sheet: /

File: Ardor_Mixer.kicad_sch

Title: Stereo buffer / mono average / AC outputs

Size: A3

Date: 2026-10-02

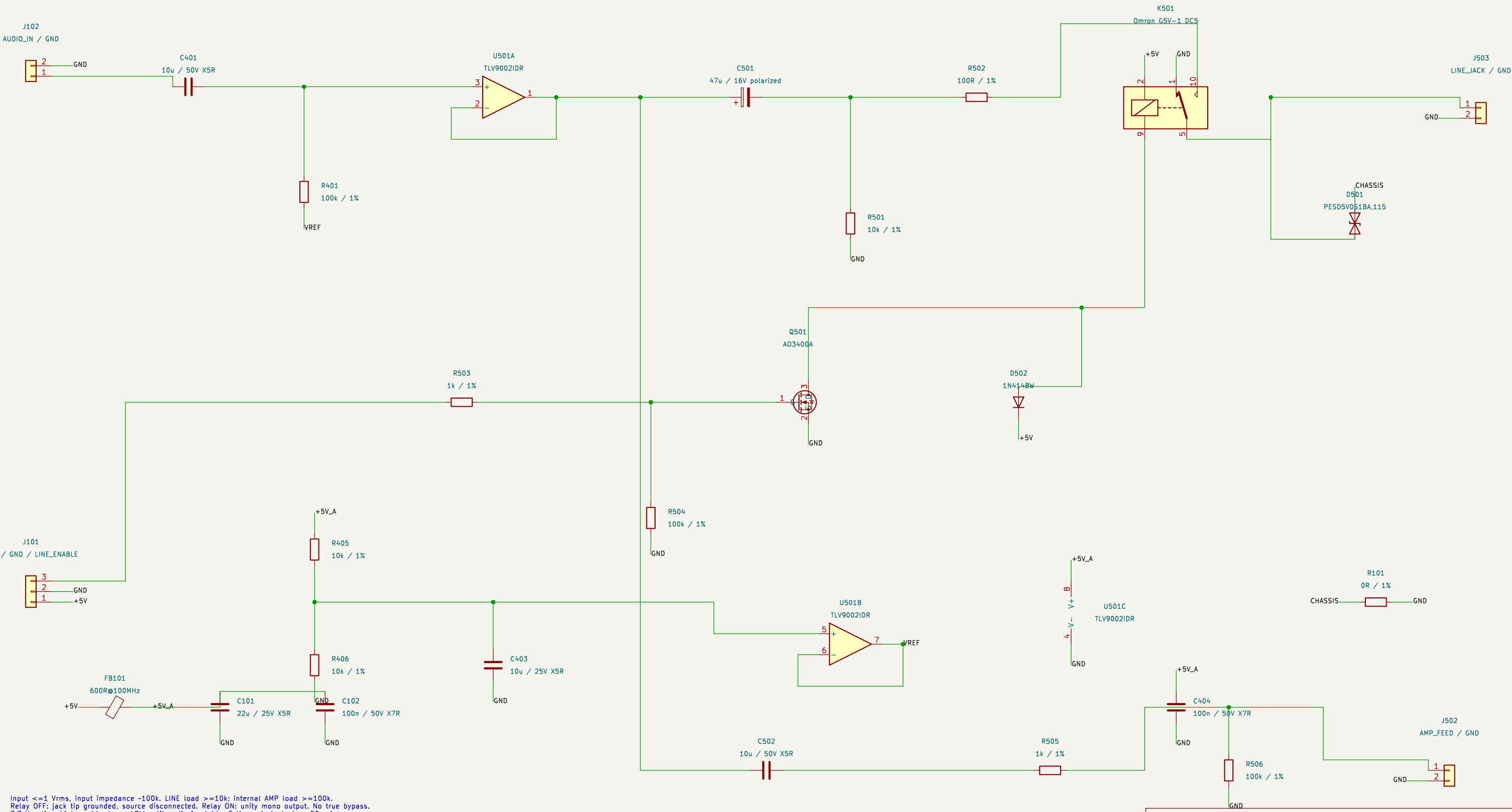
Rev: M1

KiCad E.D.A. 9.0.2

Id: 1/1

Mono line output / relay mute / internal amp feed

REV M1 | EXTERNAL HOST + REGULATED POWER | NO OTHER MODULE REQUIRED



Input <=1 Vrms, input impedance ~100k. LINE load >=10k; internal AMP load >=100k.
Relay OFF: jack tip grounded, source disconnected. Relay ON: unity mono output. No true bypass.
3.3 V active-high enable: leave LOW until audio is stable. 5 V supply budget >=50 mA.
C501 + toward buffer. J502 is a signal feed, not a speaker driver; jack sleeves bond the enclosure.

Engineering prototype – bench validation required

Arдор

Sheet: /

File: Arдор_Line_Out.kicad_sch

Title: Mono line output / relay mute / internal amp feed

Size: A3 Date: 2026-10-02

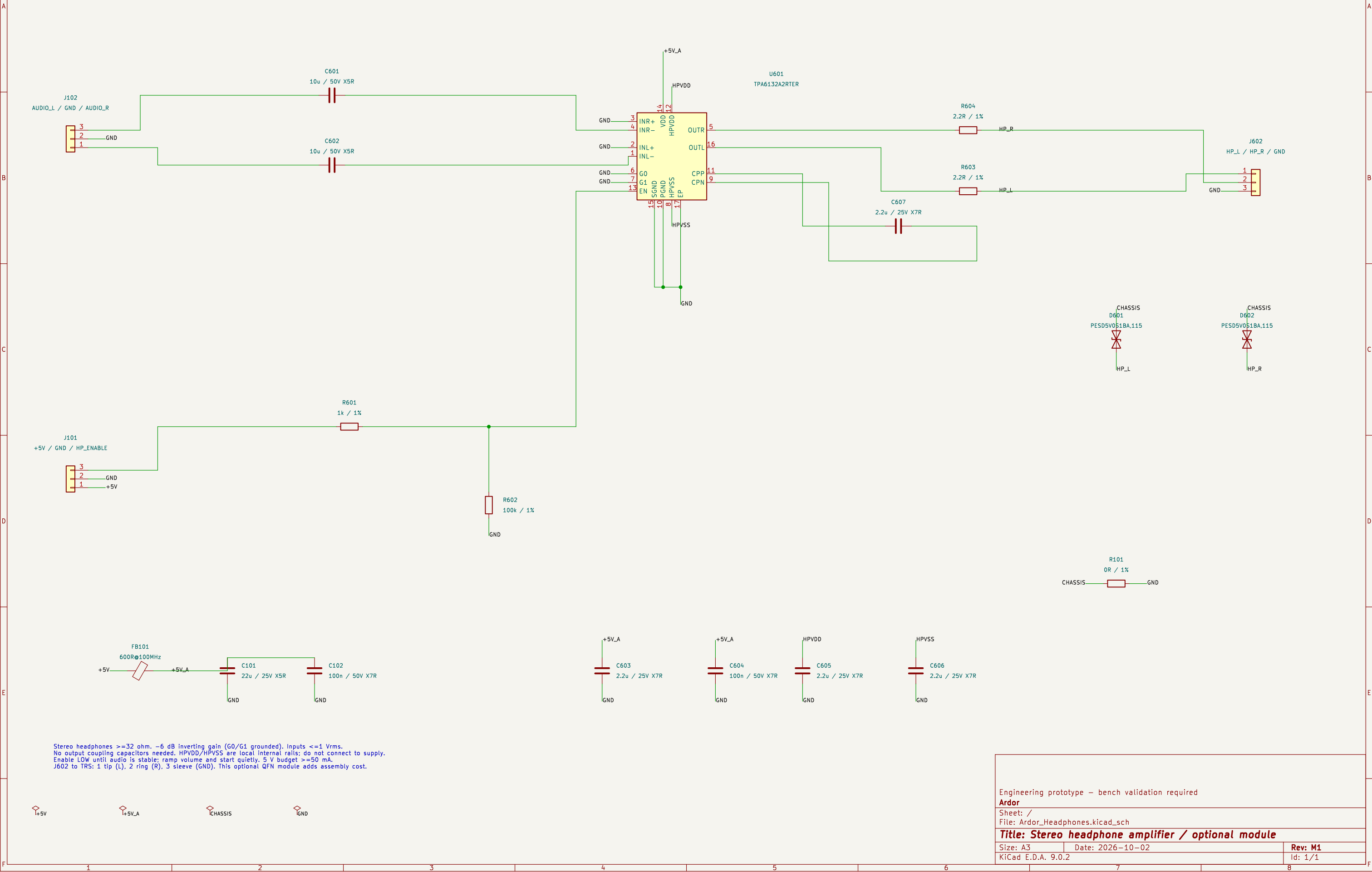
Rev: M1

KiCad E.D.A. 9.0.2

Id: 1/1

Stereo headphone amplifier / optional module

REV M1 | EXTERNAL HOST + REGULATED POWER | NO OTHER MODULE REQUIRED



Stereo headphones $\geq 32\ \Omega$. $-6\ \text{dB}$ inverting gain (G0/G1 grounded). Inputs $\leq 1\ \text{V}_{\text{rms}}$.
No output coupling capacitors needed. HPVDD/HPVSS are local internal rails; do not connect to supply.
Enable LOW until audio is stable; ramp volume and start quietly. 5 V budget $\geq 50\ \text{mA}$.
J602 to TRS: 1 tip (L), 2 ring (R), 3 sleeve (GND). This optional QFN module adds assembly cost.

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Headphones.kicad_sch

Title: Stereo headphone amplifier / optional module

Size: A3
KiCad E.D.A. 9.0.2

Date: 2026-10-02

Rev: M1

Id: 1/1